

Robson Christie

Curriculum Vitae

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PROFILE

Researcher in theoretical and computational physics developing open quantum systems approaches to problems in cosmology, quantum gravity, and astrophysics. My work combines Lindblad dynamics, stochastic unravellings, decoherence theory, quantum field theory, and numerical simulation to understand how environmental interactions shape the behavior of quantum fields in gravitational settings. I am particularly interested in using open-system methods to connect microscopic quantum dynamics with observable signatures in the early universe and high-energy astrophysical environments, and in building reproducible computational tools that turn these questions into testable calculations.

EMPLOYMENT HISTORY

SEPT 2026–PRESENT (FT)

Pusan National University (PNU)
Postdoctoral Researcher

Developing quantum mechanical models of the early universe and black holes, with a focus on Wheeler–DeWitt quantum gravity.

APR 2026–SEPT 2026 (FREELANCE)

Mercor
Physicist

Designed original physics problems that challenged frontier AI models for leading AI labs and exposed weaknesses in scientific reasoning.

MARCH 2024–MARCH 2026 (FT)

University of Portsmouth
Research Fellow

Researched open quantum systems across biology, quantum computing, and cosmology, producing publications and simulations of dissipative and inflationary dynamics.

MARCH 2023–MARCH 2024 (FT)

Imperial College London
Postdoctoral Researcher

On EPSRC grant EP/W522673/1, developed efficient simulations of quantum molecular dynamics using rare-event algorithms and estimators for transition rates.

SEPT 2018–MARCH 2023 (FT)

Imperial College London
Graduate student (PhD awarded March 2023)

Developed analytical and numerical methods for Lindblad dynamics and stochastic unravellings, studying Gaussian states and semiclassical limits of open quantum systems.

PROGRAMMING SKILLS

SCIENTIFIC COMPUTING	Python (advanced), Matlab (advanced), Mathematica (advanced), C++ (novice), , CUDA (novice)
QUANTUM COMPUTING	IBM Qiskit (advanced), QuTiP (advanced)
HPC & WORKFLOW	Git/GitHub, Slurm, PBS
SCIENTIFIC TOOLS	L ^A T _E X, ORCA, TeraChem, Quartus, LTSpice

EDUCATION

- **PhD** (Mathematics) — Imperial College London
Advisor: Dr Eva Maria Graefe
Sept 2018 – March 2023
- **MSc** (Quantum Fields and Fundamental Forces) — Imperial College London
Advisor: Prof Jonathan Halliwell
Sept 2017 – Sept 2018
- **BMat&Phys** (First class honours) — The University of Manchester
Sept 2014 – July 2017
- **A Levels** — Maths A*, Physics A*, Chemistry A*, Further Maths A, Biology A
2012 – 2014

Minkowski Kart: Relativistic Racing

Personal project

github.com/rchristie95/MinkowskiKart

Building a relativistic racing-game by forking SuperTuxKart and integrating ideas from OpenRelativity to simulate visual and gameplay effects inspired by special and general relativity. Implemented custom GLSL shader effects and C++ gameplay logic for nine relativistic power-ups, including black holes, time-dilation, cosmic strings, and Lorentz-grid frame shifts. This is a fun way for me to learn C++ and the first working version is available to download from my github: [MinkowskiKart](#).

PUBLICATIONS/PREPRINTS

Christie, R., Joo, J., Kaplanek, G., Vennin, V., Wands, D. (in preparation). Quantum Stochastic Inflation, *Manuscript in preparation*

Summary: Formulates stochastic inflation in an open-quantum-system framework, deriving a GKLS dynamics for coarse-grained scalar field variables and showing how Hubble friction and diffusion arise from the same quantum channel while recovering classical stochastic-inflation Fokker–Planck dynamics in the light-field regime.

Christie, R., Murray, C., Kim, Y., Joo, J. (2026). Non-Equilibrium Dynamics of the Time-Dependent Excitonic Coupling in Fluorescent Protein Dimers, *Journal of Physical Chemistry B* (accepted); [arXiv:2605.00027](#)

Summary: Combined transition-density coupling calculations with stochastic quantum simulations to quantify strong near-field excitonic coupling in dimeric Venus fluorescent protein and model the post-absorption transition from delocalised excitation to incoherent hopping.

Christie, R., Joo, J., Kaplanek, G., Vennin, V., Wands, D. (2025). Cosmic Lockdown: When Decoherence Saves the Universe from Tunneling, *Journal of Cosmology and Astroparticle Physics* (JCAP, accepted; selected for SISSA Medialab release); [arXiv:2512.14204](#)

Summary: Lattice simulations of a coarse-grained inflationary field show how non-adiabatic dynamics can populate false vacua; decoherence sourced by heavy modes then suppresses subsequent tunnelling, dynamically stabilising metastable states.

Christie, R., Limmer, D.T., Bolhuis, P.G. (2025). Transition Path and Interface Sampling of Stochastic Schrödinger Dynamics, *The Journal of Chemical Physics*, vol. 162, no. 11, p. 114120, 2025; [arXiv:2411.00490](#)

Summary: Extended path $\mathcal{E}$ interface sampling to quantum trajectories, enabling efficient estimation of transition rates and revealing non-Arrhenius behaviour at low temperature linked to measurement backaction (anti-Zeno effects).

Christie, R., Baek, K., Bang, J., Joo, J. (2024). Quantum Simulation of Dynamical Transition Rates in Open Quantum Systems, *arXiv preprint*, [arXiv:2412.17229](#)

Summary: Developed a quantum-simulation protocol to estimate open-system transition rates without full Lindblad evolution by decomposing rates into independently measurable contributions; validated on IBM quantum hardware.

Christie, R., Trayford, J. (2024). The Sound of Decoherence, **Sonification Award Winner 2025**, [arXiv:2412.17045](#); [sonification.design/Decoherence](#)

Summary: Created an audio mapping from density-matrix structure to perceptual features (energy levels as tones, coherences as binaural cues), enabling listeners to track tunnelling, thermalisation, decoherence, and recoherence in Lindblad dynamics.

Christie, R., Eastman, J. (2024). Quantum Tunnelling and Thermally Driven Transitions in a Double Well Potential at Finite Temperature, *Journal of Physics A: Mathematical and Theoretical*, vol. 57, no. 23, p. 235005, 2024; [arXiv:2312.12101](#)

Summary: Mapped the thermal-to-quantum tunnelling crossover in a dissipative double well, showing how low-temperature noise can randomise yet preserve tunnelling, and introduced a Lindblad Brownian-motion model that removes the zero-temperature singularity.

Christie, R., Eastman, J., Schubert, R., Graefe, E.M. (2022). Quantum-jump vs stochastic Schrödinger dynamics for Gaussian states with quadratic Hamiltonians and linear Lindbladians. *Journal of Physics A: Mathematical and Theoretical*, vol. 55, no. 45, p. 455302, 2022; [arXiv:2203.11530](#)

Summary: Compared continuous (stochastic Schrödinger) and jump unravellings of Lindblad dynamics, showing identical ensemble limits but qualitatively different single-trajectory behaviour. Derived closed-form Gaussian evolution for continuous trajectories and introduced a Gaussian Hagedorn-wavepacket scheme for jump trajectories, validated on measured and damped oscillators.